

TOWARDS PERCEPTUAL SOUNDSCAPE CHARACTERIZATION USING EVENT DETECTION ALGORITHMS

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ABSTRACT

Assessing properties about specific sound sources is important to characterize better the perception of urban sound environments. In order to produce perceptually motivated noise maps, we argue that it is possible to consider the data produced by acoustic sensor networks to gather information about sources of interest and predict their perceptual attributes.

To validate this important assumption, this paper reports on a perceptual test on simulated sound scenes for which both perceptual and acoustic source properties are known. Results show that it is indeed feasible to predict perceptual source-specific quantities of interest from recordings, leading to the introduction of two predictors of perceptual judgments from acoustic data. The use of those predictors in the new task of automatic soundscape characterization is finally discussed.

Index Terms— Soundscape, urban acoustic monitoring, event detection

1. INTRODUCTION

The ongoing urbanization process has led to an increase in sound quality concerns. In urban areas the noise has been linked to several health issues including sleep-related troubles as well as heart disease rates, and is a major cause for city dwellers' annoyance in certain areas. In this context, the 2002/49/CE European directive [1] requires that large cities maintain noise maps to facilitate the development of noise reducing plans. These noise maps are mainly based on predictive maps generated using propagation and emission acoustic models. The studies are also 1) often limited to traffic and other transportation sources, and 2) no fusions of simulations with physical measurements are used. Furthermore, the models depend on data that may be at times or in certain locations unavailable or incomplete. The advent of the internet of things (IoT) presents an opportunity for the development of large, scalable networks of acoustic sensors [2, 3]. The "characterization of urban sound environments" (CENSE) project [4] aims at implementing such a network to produce perceptually motivated noise maps.

The ISO 12913-1 [5] standard gives the following definition of soundscape: "the acoustic environment as perceived and understood and/or experienced by people and/or society, in context". The assessment of subjective descriptors [6, 7, 8] such as the liveliness or calmness is thus necessary to evaluate the quality of urban scenes. The relevant attributes describing the appreciation of soundscapes can be mapped in perceptual spaces [9, 10]. The set of considered attributes is reduced to a few dimensions which are used as a basis

for perceptual experiments. Specifically, the dimension of pleasantness is increasingly associated with soundscape quality in recent works [11, 12, 13, 14]. Soundscape perception is highly dependent on the composition of the scene [15, 16]. Indeed, each sound source yields a different perceptual response. For example, soundscape pleasantness is likely to be improved by birdsongs and deteriorated by mechanical noises.

Acoustic monitoring applications typically rely on the measurement of energetic (sound levels, eg. L_{Aeq}) and psychoacoustic (eg. Zwicker's loudness N) indicators. These global quantities describe the overall activity, with percentile values linked to event or background assessment. However they do not differentiate sound sources and are thus not sufficient to a perceptual characterization of soundscapes. Additional information about the taxonomic classification of active sources and their distribution in time is needed. Several sets of relevant indicators have been studied [17, 18, 19] to better account for the specificities of each scene and their source composition.

The use of large-scale sensor networks yields a problematic for the extraction of content-related quantities of interest from important amounts of data. Despite a growing interest in the community, machine learning models - to the best of our knowledge - were not yet specifically targeted to the prediction of source-specific perceptual parameters in complex urban environments. Most event detection applications focus on obtaining a precise annotation of source activity, within usual ranges of tens of milliseconds. The estimation of sound levels involves entirely different models through source separation and regression [20] and longer time scales.

We believe that the use of machine listening techniques could greatly benefit the automatic assessment of urban soundscape quality using sensor networks. The aim of this paper is to 1) bring some context of soundscape characterization, and 2) report on a perceptual experiment performed in order to study which features shall be brought by automatic event detection systems in order to gather relevant information for the task of characterize perceptual attributes of the soundscape.

2. SOUNDSCAPE CHARACTERIZATION

Urban soundscape monitoring has only scarcely been studied by the machine listening community [21]. This work aims at contributing to this task by focusing on pleasantness as it is the most recurrent descriptor of urban soundscape quality, though similar studies could be led for other notions such as liveliness.

Several perceptual experiments on the urban soundscape quality have indeed proposed a model of pleasantness from other per-

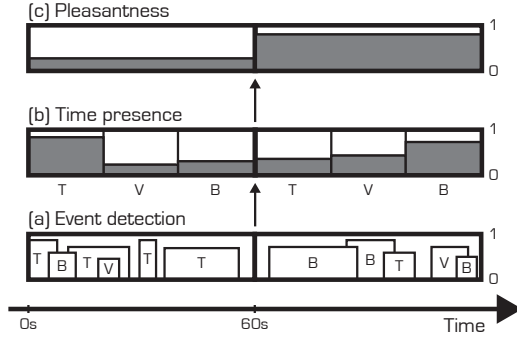


Figure 1: The three suggested levels of metrics to predict soundscape pleasantness. (a) Traffic (T), voice (V) and bird (B) events are detected and their sound level roughly estimated. (b) The perceptual time of presence for each source is computed on one-minute frames, resulting in a pleasantness value (c).

ceptual parameters [22, 9, 14, 13]. In all cases, a good approximation of pleasantness can be obtained by linear combination of both overall and source-specific parameters evaluated on discrete scales. Global parameters consider the sound scene in its entirety for which the overall loudness is commonly used. The parameters used for the assessment of source-wise contributions include 1) the sound level where each source is considered separately, 2) the emergence or dominance relating to the influence of the source in the global mix, or 3) the time of presence, that is the ratio of time where the source s is heard in a given scene. The notion of time of presence is of particular interest as it hints at the possibility of automatic prediction through event detection systems. The corresponding model is:

$$P = aL + \sum_s b_s T_{s,p} + c \quad (1)$$

where P is the scene's pleasantness, L is the perceived overall level and $T_{s,p}$ is the perceived time of presence for source s . These parameters are evaluated on discrete scales through perceptual tests. The coefficients a , b_s and c are usually found via multiple linear regression and thus differ in each study. Furthermore, three principal source categories are usually identified: mechanical, human and nature. Mechanical sounds are mainly composed of traffic and are mostly found to have a negative impact on soundscape pleasantness, whereas nature sources such as bird activity or water sounds have a positive influence and human sounds (voices) can yield mixed effects.

Assuming this perceptual model, the prediction of pleasantness can be assimilated as that of perceived times of presence of sources. Three levels of metrics are thus identified. First, the physical level Figure 1(a) is evaluated on the presence and emergence of the three identified sound sources: traffic (T), voice (V) and birds (B). The second level Figure 1(b) is the perceived time of presence for each source represented as a scalar in the 0-1 range. The third level Figure 1(c) is the estimate of pleasantness, also represented as a 0-1 scalar. Both the perceptual levels of metrics are only relevant on longer time scales, about one minute being a usual value in existing experiments.

The transition model between the perceived time of presence per source and pleasantness has already been proposed. However no previous work exists that uses detection models for the estimation of source-specific subjective parameters. The feasibility of assessing

source perception from the postulated metrics at the physical level shall be verified as a first step prior to building the full estimation model.

3. FROM PHYSICAL TO PERCEPTUAL TIME OF PRESENCE OF SOURCES

We thus conduct a perceptual experiment to validate this key step of the estimation procedure. We wish to study the relation between extracted source-dependent physical indicators to their perceptual equivalents, then validate the relevance of the first level of metrics introduced in the previous section.

3.1. Perceptual Test

For this test, a set of sound scenes recorded in the 13th district of Paris as part of the GRAFIC project [14] is used as reference. Some artificial scenes with equivalent event sequencing are also used for which the acoustic properties of each active source can be computed precisely.

Of the 19 different recording locations, 9 are selected to represent diverse compositional properties: park (P3, P9), quiet street (P5, P11, P13, P17), noisy street (P2, P6) and very noisy street (P16). Corresponding artificial scenes are simulated following the method described in [23]. Simulations are obtained using the *sim-Scene* software [24]. To do so, the recordings are first annotated by identifying active background and event sources. Background sounds are present throughout the whole scene and are characterized by an absolute level parameter. Conversely, events are localized occurrences that are defined by their onset and duration as well as an event-to-background ratio (EBR). The sound scenes are simulated from these annotations and a database of extracts for isolated sources obtained on *freesound.org*, see [23] for more details. This ensures that ground truth source-specific presence and sound level can be computed. One minute of audio is extracted for each scene such as no single event overwhelms the rest of the excerpt.

During the test, the order of appearance is as follows: the original recorded scenes from locations P3 and P16 representing quiet and very noisy environments are always presented first to help participants use the full range of the scale during the test. The 9 simulated sounds are then presented in random order to limit order biases over the participants population. For each scene, 14 criteria are evaluated on a 0-10 scale by the subject. These parameters are displayed in French, but translated in English in this paper for the sake of clarity. The first four questions cover global perceptual parameters:

1. *Noisy - Quiet*: Overall perceived loudness (OL),
2. *Boring, uninteresting - Stimulating, interesting*: Interest (I),
3. *Inert, amorphous - Lively, eventful*: Liveliness (L),
4. *Agitated, chaotic - Calm, peaceful*: Calmness (C).

Source-specific perceived time of presence (scale *Never - Continuously*) and sound level (scale *Very low - Very high*) are also evaluated. The considered sources are traffic (T), birds (B), horns and sirens (H), human voice (V) and footsteps (F). The perceived time of presence and level for source s are respectively noted $T_{s,p}$ and $L_{s,p}$ in the remainder of this paper.

Participants can only listen to each scene once and must answer all questions before proceeding to the next scene. All subjects used the same hardware desktop configuration, sound card and software,

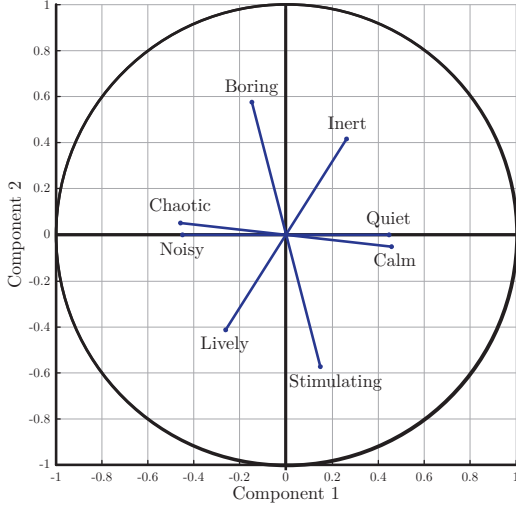


Figure 2: Principal component analysis (first two components) of the four general perceptual parameters at the scene level ($n=9$). The observed space is distorted although comparable that of previous works in the literature.

as well as Beyerdynamics DT-990 headphones in a quiet environment. The same output volume on headphones was set by the experimenter for all scenes and participants. The resulting playback sound level ranged from approximately 50 dB to 78 dB over the corpus. 30 subjects took the test in 3 sessions, all reported normal hearing conditions.

3.2. Perceptual space

An outlier detection procedure is applied on the 270 resulting assessments (30 subjects, 9 scenes). An assessment is rejected when its distance from the mean is higher than 3 standard deviations at the question level, that is for each parameter of each scene. The results from two participants with more than 10% assessments considered as outliers are removed from the study.

The perceptual space produced by the test is first compared to previous studies in the literature. This is to ensure that relevant conclusions can be made on further analysis. Figure 2 shows the standardized principal component analysis (PCA) of the average values of the four general questions at the scene level ($n=9$). The first two components respectively explain 52.3% and 30.5% of the global variance. It is found that liveliness (L) correlates poorly with calmness (C), while interest (I) is between the two. These results can be compared to previous studies on similar soundscape qualities parameters [9, 10, 25], where interest and calmness were established as almost independent. The scale of liveliness was in both cases correlated similarly with the two others. However, just as the principal components space is slightly distorted in [25] due to the study being focused on park environments. The correspondence between these results on global perceptual parameters and those of the literature allows us to think that perceptual data on sources are relevant for this study.

3.3. Proposed indicators

As discussed in Section 2 several models have been established to assess pleasantness as a function of global and source-specific parameters. The main objective of this work is to link physical indicators to perceptual source-specific parameters to ultimately predict pleasantness from acoustical data without perceptual assessments. Thus, physical indicators are computed from the audio tracks obtained during scene simulation. To evaluate the overall loudness of the scene, three measurements are chosen in accordance to previous studies [11, 13, 14]:

- L_{50} : Z-weighted (no weighting over the observed frequency range) sound level exceeded 50% of the time in dB,
- L_{A50} : A-weighted sound level exceeded 50% of the time in dBA,
- L_{50} for the 1kHz band only.

Source-specific indicators are also computed: the time of presence and an emergence estimation metric (resp. T_s and L_s for source s), obtained by subtracting the global L_{90} (Z-weighted level exceeded 90% of the time), found to represent well background activity, to the L_{10} of each source. Sound levels are computed with the Matlab ITA toolbox [26] in the 20 Hz-20 kHz range.

In the considered scenes, background sources are always active. The measurement of time of presence is thus limited to sound events which leads to relatively poor representation of the scene perception. Furthermore, the ground truth indicators are computed for each source separately and do not consider the potential impact of other sources active at the same time. Two additional indicators are thus designed regarding these considerations.

The first proposed indicator $T_s(\alpha)$ is a time of presence metric relying on the emergence of each sound source relative to the others. Sound levels (dB) are computed for audio frames of 125 ms. This duration is approximately that of the shortest event found during annotation and corresponds to the "fast" measurements used in acoustical monitoring applications. The emergence, *i.e.* difference $\Delta_s(t)$ of sound levels between the studied source ($L_s(t)$) and the background constituted of all others ($L_b(t)$) is computed. The source is then considered present on a given time frame if the emergence is greater than a threshold value α . A time of presence measurement is obtained by averaging over time:

$$T_s(\alpha) = \frac{1}{N_t} \sum_{t=1}^{N_t} \mathbb{1}_{\Delta_s(t) > \alpha} \quad (2)$$

where N_t is the total number of 125 ms analysis frames in the scene. The optimal threshold α is optimized via grid search to maximize the resulting correlation with the 45 average perceptual time of presence assessments. An optimal value of $\alpha = -31\text{dB}$ for the considered corpus is found. As the sound levels of tested scenes range from 50 dB to 78 dB (cf. Section 3.1), only sources with very low sound level on the whole spectrum are considered not heard.

However, the masking of a sound by another does not depend only on the emergence over the whole frequency spectrum. The spectral distribution is important, the level comparison shall thus be made around the characteristic frequency components of a source. A second indicator $T_s(\alpha, \beta)$, based on a spectral decomposition is thus proposed. Third-octave bands sound levels are computed on 125 ms frames and the emergence of a source compared to the background is defined as

$$\Delta_s(t, f) = L_s(t, f) - L_b(t, f) \quad (3)$$

Table 1: Pearson correlation coefficients between perceptual parameters and physical indicators at the scene level (n=9). *: $p < 0.05$, **: $p < 0.01$, non-significant correlations ($p > 0.05$) are noted NS.

Phys./Perc.	OL	I	L	C	$L_{T,p}$	$T_{T,p}$	$L_{B,p}$	$T_{B,p}$	$L_{H,p}$	$T_{H,p}$	$L_{V,p}$	$T_{V,p}$	$L_{F,p}$	$T_{F,p}$
$L_{50,1kHz}$	0.93**	NS	NS	-0.92**	0.75*	0.7*	NS	NS	NS	NS	NS	NS	NS	NS
L_{50}	0.98**	NS	0.73*	-0.97**	0.72*	NS	NS	NS	NS	NS	NS	NS	NS	NS
L_{A50}	0.96**	NS	0.73*	-0.94**	NS	NS	NS	NS	NS	NS	NS	NS	NS	NS
T_T	NS	NS	NS	NS	NS	NS	NS	NS	NS	NS	NS	NS	NS	NS
L_T	NS	NS	NS	NS	NS	NS	NS	NS	NS	NS	NS	NS	NS	NS
T_B	NS	0.67*	NS	NS	0.71*	0.75*	NS	NS	NS	NS	NS	NS	NS	NS
L_B	NS	0.93**	NS	NS	-0.84**	-0.83**	0.91**	0.82**	NS	NS	NS	NS	NS	NS
T_H	NS	NS	NS	NS	NS	NS	NS	NS	NS	0.84**	NS	NS	NS	NS
L_H	NS	NS	NS	NS	NS	NS	NS	NS	0.98**	0.78*	NS	NS	NS	NS
T_V	NS	NS	NS	NS	NS	NS	NS	NS	NS	NS	NS	NS	NS	NS
L_V	NS	NS	0.81**	NS	NS	NS	NS	NS	NS	NS	0.84**	0.88**	NS	NS
T_F	NS	NS	NS	NS	NS	NS	NS	NS	NS	NS	NS	NS	0.9**	0.68*
L_F	NS	NS	-0.72*	NS	NS	NS	NS	NS	NS	NS	-0.69*	-0.78*	0.92**	NS
$T_T(\alpha)$	NS	-0.81**	NS	NS	0.90**	0.94**	NS	NS	NS	NS	NS	NS	NS	NS
$T_T(\alpha, \beta)$	NS	-0.80**	NS	NS	0.88**	0.92**	NS	NS	NS	NS	NS	NS	NS	NS
$T_B(\alpha)$	NS	0.88**	NS	NS	NS	NS	0.95**	0.97**	NS	NS	NS	NS	NS	NS
$T_B(\alpha, \beta)$	NS	0.88**	NS	NS	NS	NS	0.95**	0.97**	NS	NS	NS	NS	NS	NS
$T_H(\alpha)$	NS	NS	NS	NS	NS	NS	NS	NS	NS	0.83**	NS	NS	NS	NS
$T_H(\alpha, \beta)$	NS	NS	NS	NS	NS	NS	NS	NS	0.73*	0.88**	NS	NS	NS	NS
$T_V(\alpha)$	NS	NS	0.82**	NS	NS	NS	NS	NS	NS	NS	0.79*	0.83**	NS	NS
$T_V(\alpha, \beta)$	NS	NS	0.82**	NS	NS	NS	NS	NS	NS	NS	0.75*	0.79*	NS	NS
$T_F(\alpha)$	NS	NS	NS	NS	NS	NS	NS	NS	NS	NS	NS	-0.71*	0.87**	NS
$T_F(\alpha, \beta)$	NS	NS	NS	NS	NS	NS	NS	NS	NS	NS	NS	NS	0.90**	0.70*

Similarly to the first metric $T_s(\alpha, \beta)$ then relies on simple thresholds applied on the emergence, first in frequency then in time. Its expression is as follows:

$$T_s(\alpha, \beta) = \frac{1}{N_t} \sum_{t=1}^{N_t} \mathbb{1} \left[\frac{\sum_{f=1}^{N_f} \Delta_s(t, f) \mathbb{1}_{\Delta_s(t, f) > \alpha}}{\sum_{f=1}^{N_f} \mathbb{1}_{\Delta_s(t, f) > \alpha}} > \beta \right] \quad (4)$$

where N_f is the number of third-octave bands. Here the emergence threshold α is applied to each frequency band of the signal at a given time frame. To determine if the source is heard in the frame a second threshold β is then used on the mean emergence of the source on emergent bands. Again, optimal values for parameters $\alpha_{opt} = -6dB$ and $\beta_{opt} = -5dB$ are found via grid search on the experiment corpus as no other subset of scenes with both physical and perceptual data is available. This set of values is more plausible physically, as it indicates that a source is considered heard if its sound level is at most 5 dB lower than that of other sources overlapping in time and frequency.

Table 1 shows the Pearson's correlation coefficients between the computed indicators and assessed parameters at the sound scene level (n=9). The three globally computed sound levels L_{50} , L_{A50} and $L_{50,1kHz}$ represent well the perceived overall loudness of the scene and can be used directly for pleasantness prediction. Ground truth emergences also correlate with the evaluated sound level parameters for all sources but traffic. The perceived time of presence is however represented poorly by its corresponding ground truth estimation in common background sources: traffic, birds and human voices. Traffic, specifically, is almost always present throughout a scene in real life conditions. Its ground truth activity thus does not vary significantly across the considered corpus, although high variations in perceptual assessments indicate that it may not be heard at all time. The two proposed indicators successfully account for this effect for background sources while yielding similar correlations for horns and footsteps. Furthermore, these parameters are more discriminative for traffic and birds. This confirms the need of an emergence-based time of presence indicator to successfully

represent heard sources in the scene's mix.

For all sources the perceived time of presence and sound level are highly correlated ($r > 0.8, p < 0.01$). This is not the case for the corresponding acoustic indicators, indicating information redundancy between these two quantities at the perceptual level. As a result one of the two quantities is often omitted in proposed pleasantness models.

4. CONCLUSION

A pilot experiment was performed to assess the relevance of predicting perceptual parameters from acoustic indicators in simulated scenes for soundscape quality assessment. The ground truth time of presence of sources is found not sufficient to fully characterize soundscape perception. Some sources can be active but not heard in the mix, especially background sounds such as traffic. This illustrates the need to design a masking model-based metric to determine each source's perceptual importance in complex soundscapes. The proposed indicator $T_s(\alpha, \beta)$, while relying on a basic emergence model due to the small amount of available data, can be directly linked to source-specific perceptual quantities. Predicting the average pleasantness of a soundscape can thus be achieved by estimating the source activity and emergence indicators proposed in Section 2.

Precision requirements of the postulated physical metrics are also obtained. 125 ms or longer time scales used for the computation of all indicators in the presented experiment allow the design of perceptually relevant indicators. A binary masking model is shown in this study to improve parameter prediction. The estimation of source-wise emergence as a classification process (e.g. 4 classes from *Not heard at all* to *Dominant*) as opposed to continuous regression is thus sufficient for the application needs.

Future work will 1) consider a refined perceptual experiment with a richer soundscape corpus in order to achieve a stronger validation and model design including comparison with state-of-the-art masking models and 2) formulate a complete experimental protocol dedicated to the soundscape characterization task.

5. REFERENCES

- [1] EC, "Directive 2002/49/ec of the european parliament and of the council of 25 june 2002 relating to the assessment and management of environmental noise," *Off. J. Eur. Communities*, vol. 189, p. 12, 2002.
- [2] C. Mydlarz, J. Salamon, and J. Bello, "The implementation of low-cost urban acoustic monitoring devices," *Applied Acoustics*, vol. 117, pp. 207–218, 2017.
- [3] F. Gontier, M. Lagrange, P. Aumond, A. Can, and C. Lavandier, "An efficient audio coding scheme for quantitative and qualitative large scale acoustic monitoring using the sensor grid approach," *Sensors*, vol. 17, 2017.
- [4] J. Picault, A. Can, J. Ardouin, P. Crepeaux, T. Dhome, D. Ecotiere, M. Lagrange, C. Lavandier, V. Mallet, C. Mitlicki, and M. Paboeuf, "Characterization of urban sound environments using a comprehensive approach combining open data, measurements, and modeling," in *Acoustics '17, Boston*, 2017.
- [5] ISO 12913-1:2014, "Acoustics - soundscape - part 1: definition and conceptual framework," International Organization for Standardization, Geneva, CH, Standard, 2014.
- [6] B. Berglund and M. Nilsson, "On a tool for measuring soundscape quality in urban residential areas," *Acta Acust. unit. Acust.*, vol. 92, pp. 938–944, 2006.
- [7] A. Brown, "Towards standardization in soundscape preference assessment," *Applied Acoustics*, vol. 72, pp. 387–392, 2011.
- [8] F. Aletta, J. Kang, and O. Axelsson, "Soundscape descriptors and a conceptual framework for developing predictive soundscape models," *Landsc. Urban Plan.*, vol. 149, pp. 65–74, 2016.
- [9] O. Axelsson, M. Nilsson, and B. Berglund, "A principal components model of soundscape perception," *J. Ac. Soc. Am.*, vol. 128, p. 2836, 2010.
- [10] R. Cain, P. Jennings, and J. Poxon, "The development and application of the emotional dimensions of a soundscape," *Applied Acoustics*, vol. 74, pp. 232–239, 2013.
- [11] B. D. Coensel and D. Botteldooren, "The quiet rural soundscape and how to characterize it," *Acta Acust. unit. Acust.*, vol. 92, pp. 887–897, 2006.
- [12] P. Delaitre, C. Lavandier, C. Ribeiro, M. Quoy, E. D'Hondt, E. G. Boix, and K. Kambona, "Influence of loudness of noise events on perceived sound quality in urban context," in *Inter Noise*, 2014.
- [13] P. Ricciardi, P. Delaitre, C. Lavandier, F. Torchia, and P. Aumond, "Sound quality indicators for urban places in paris cross-validated by milan data," *J. Ac. Soc. Am.*, vol. 138, pp. 2337–2348, 2014.
- [14] P. Aumond, A. Can, B. D. Coensel, D. Botteldooren, C. Ribeiro, and C. Lavandier, "Modeling soundscape pleasantness using perceptive assessments and acoustic measurements along paths in urban context," *Acta Acust. unit. Acust.*, vol. 103, pp. 430–443, 2017.
- [15] C. Lavandier and B. Defreville, "The contribution of sound source characteristics in the assessment of urban soundscapes," *Acta Acust. unit. Acust.*, vol. 92, pp. 912–921, 2006.
- [16] M. Nilsson and B. Berglund, "Soundscape quality in suburban green areas and city parks," *Acta Acust. unit. Acust.*, vol. 92, pp. 903–911, 2006.
- [17] A. Can, L. Leclercq, J. Lelong, and J. Defrance, "Capturing urban traffic noise dynamics through relevant descriptors," *Applied Acoustics*, vol. 69, pp. 1270–1280, 2008.
- [18] A. Can, P. Aumond, S. Michel, B. D. Coensel, C. Ribeiro, D. Botteldooren, and C. Lavandier, "Comparison of noise indicators in an urban context," in *45th International Congress and Exposition of Noise Control Engineering*, 2014.
- [19] L. Brocolini, C. Lavandier, M. Quoy, and C. Ribeiro, "Measurement of acoustic environments for urban soundscapes: choice of homogeneous periods, optimization of durations, and selection of indicators," *J. Ac. Soc. Am.*, vol. 134, pp. 813–821, 2013.
- [20] J. Gloaguen, A. Can, M. Lagrange, and J. Petiot, "Estimating traffic noise levels using acoustic monitoring: a preliminary study," in *DCASE 2016, Detection and Classification of Acoustic Scenes and Events*, 2016.
- [21] J. P. Bello, C. Mydlarz, and J. Salamon, "Sound analysis in smart cities," in *Computational Analysis of Sound Scenes and Events*, T. Virtanen, M. D. Plumbley, and D. P. W. Ellis, Eds. Springer International Publishing, 2018, pp. 373–397.
- [22] M. Nilsson, D. Botteldooren, and B. D. Coensel, "Acoustic indicators of soundscape quality and noise annoyance in outdoor urban areas," in *19th International Congress on Acoustics*, 2007.
- [23] J. Gloaguen, A. Can, M. Lagrange, and J. Petiot, "Creation of a corpus of realistic urban sound scenes with controlled acoustic properties," in *Proceedings of Meetings on Acoustics*, 2017.
- [24] G. Lafay, M. Lagrange, M. Rossignol, E. Benetos, and A. Roebel, "A morphological model for simulating acoustic scenes and its application to sound event detection," *IEEE/ACM Transactions on Audio, Speech and Language Processing*, vol. 24, no. 10, pp. 1854–1864, 2016. [Online]. Available: <https://hal.archives-ouvertes.fr/hal-01111381>
- [25] J. Jeon, J. Hong, C. Lavandier, J. Lafon, O. Axelsson, and M. Hurtig, "A cross-national comparison in assessment of urban park soundscapes in france, korea, and sweden through laboratory experiments," *Applied Acoustics*, vol. 133, pp. 107–117, 2018.
- [26] M. Berzborn, R. Bomhardt, J. Klein, J. Richter, and M. Vorlander, "The ita-toolbox: an open source matlab toolbox for acoustic measurements and signal processing," in *43th Annual German Congress on Acoustics*, 2017.